

Pressure

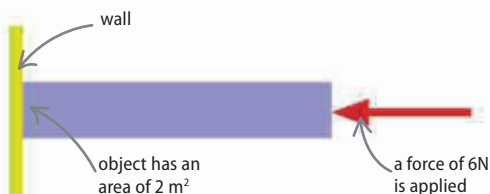
PRESSURE IS THE RESULT OF ONE THING PRESSING ON THE SURFACE OF ANOTHER.

Pressure can be applied to or by any medium, including air and water.

What is pressure?

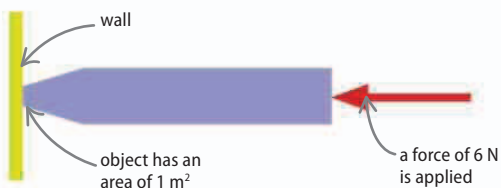
Pressure is defined as force per unit area, and is measured in pascals (Pa), which is equal to one newton per square meter. To calculate the pressure, divide the force pressing on the object by the area it is spread across.

$$\text{Pressure (pascals (Pa))} = \frac{\text{Force (newtons (N))}}{\text{Area (m}^2\text{)}}$$



△ Larger area, lower pressure

If a force of 6 N is spread over an area of 2 m², the applied pressure is 6 divided by 2, which equals 3 Pa.



△ Smaller area, higher pressure

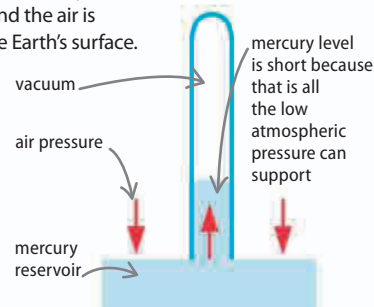
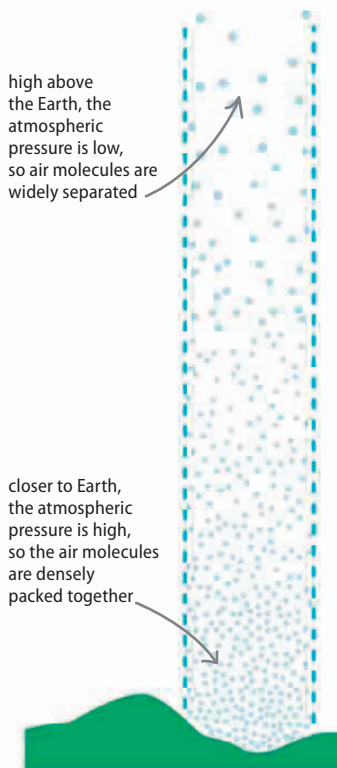
If a force of 6 N is spread over an area of 1 m², the applied pressure is 6 divided by 1, which equals 6 Pa. This is why drawing pins and nails are pointed: their small-area tips apply very high pressures, so they penetrate materials easily.

Atmospheric pressure

At the Earth's surface, the atmosphere applies a pressure of about 101,000 pascals to all objects. We cannot feel this pressure because it is balanced by an equal and opposite pressure inside our bodies. In different weather conditions and at different heights, the local atmospheric pressure changes. This can be measured using a barometer.

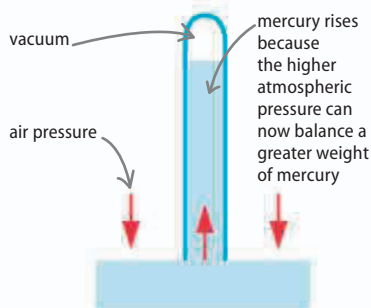
▽ Height and pressure

Gas molecules are constantly on the move and bumping into each other. When they hit another molecule or the wall of a container, they exert pressure on it. The air molecules close to the Earth are at the bottom of the atmosphere with all the other air molecules on top of them. Therefore, the pressure is higher and the air is denser. Air grows thinner higher from the Earth's surface.



△ Low atmospheric pressure

If the air pressure above the reservoir is low, it will not produce enough force to make the mercury rise up the tube.



△ High atmospheric pressure

If the air pressure above the reservoir is high, it will push the mercury up the tube.

SEE ALSO

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Weather

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